

Why do Students Avoid Each Other?

Investigating Positive and Negative Network Effects

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 - Broad literature on these (e.g., Huitsing et al., 2014, among others)
 - Less studied
- Avoidance is particularly interesting since it is often more frequent and less socially complex than bullying and aggression (Wójcik and Flak, 2019; Kros et al., 2021)

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- (1) Structural dependencies in classroom avoidance networks
- (2) How a student's tendency to avoid others and be avoided relates to their position in the classroom friendship network

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- (3) Use a larger sample of classroom networks

Participants and procedures

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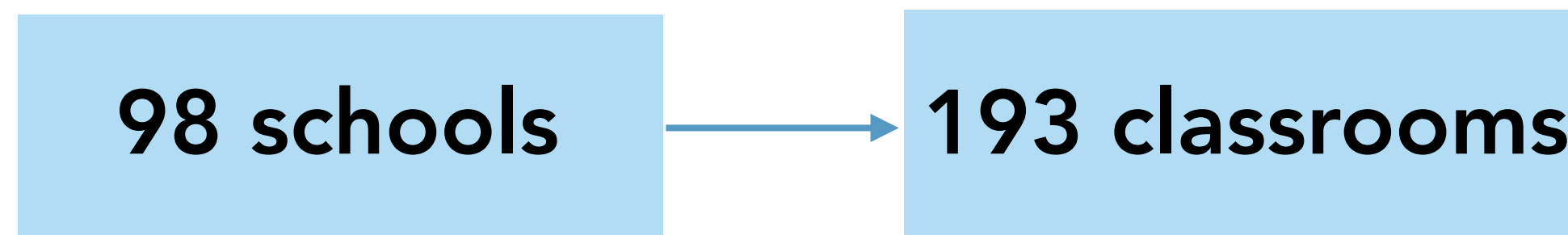
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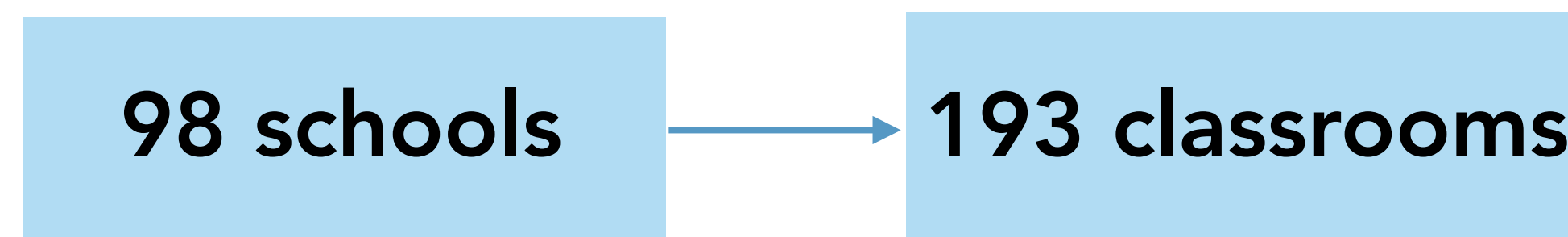


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Student i nominates classmate j as a friend if they nominate them for (1) or (2)

Some students did not complete all the self-reported social relationship questions

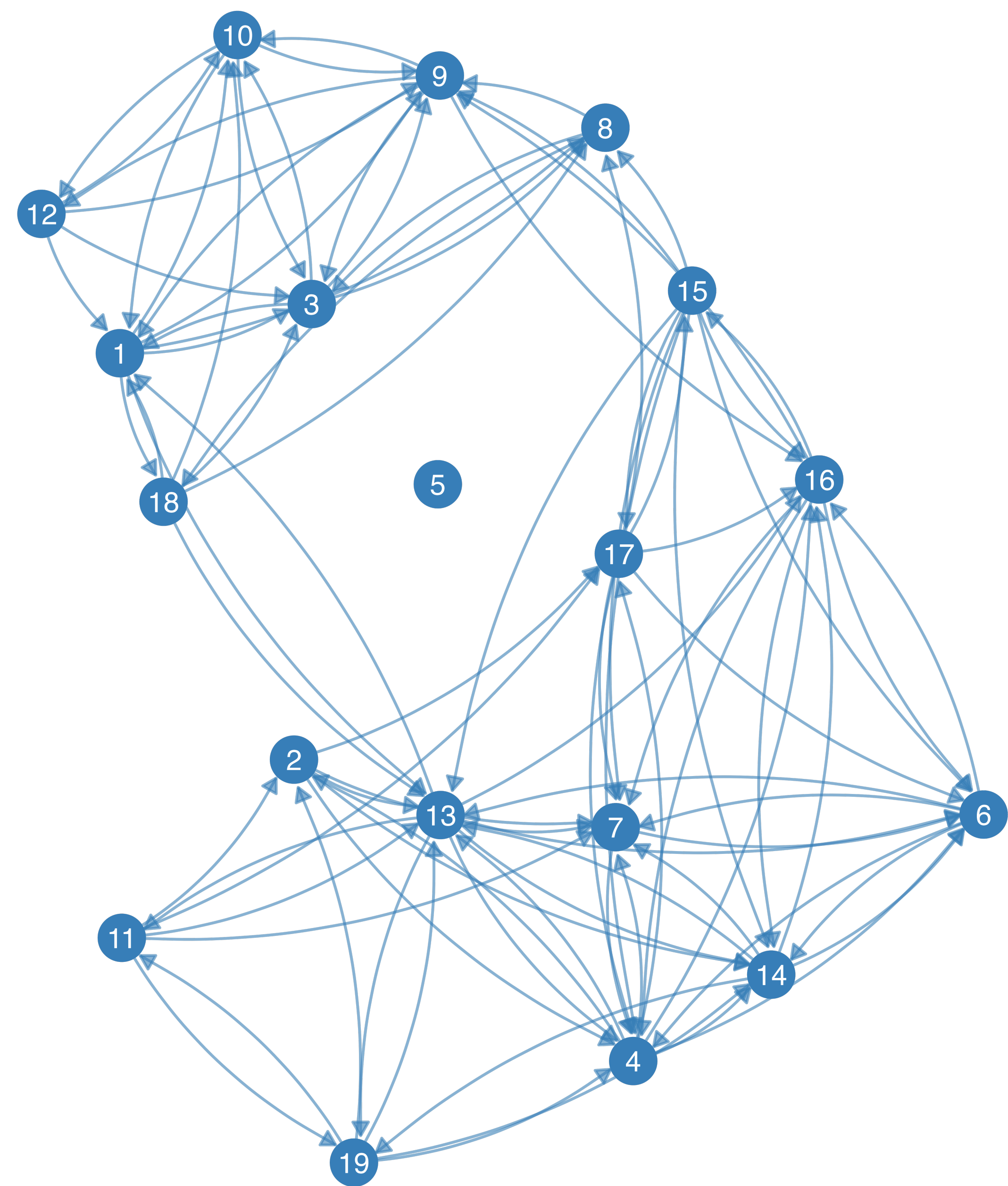
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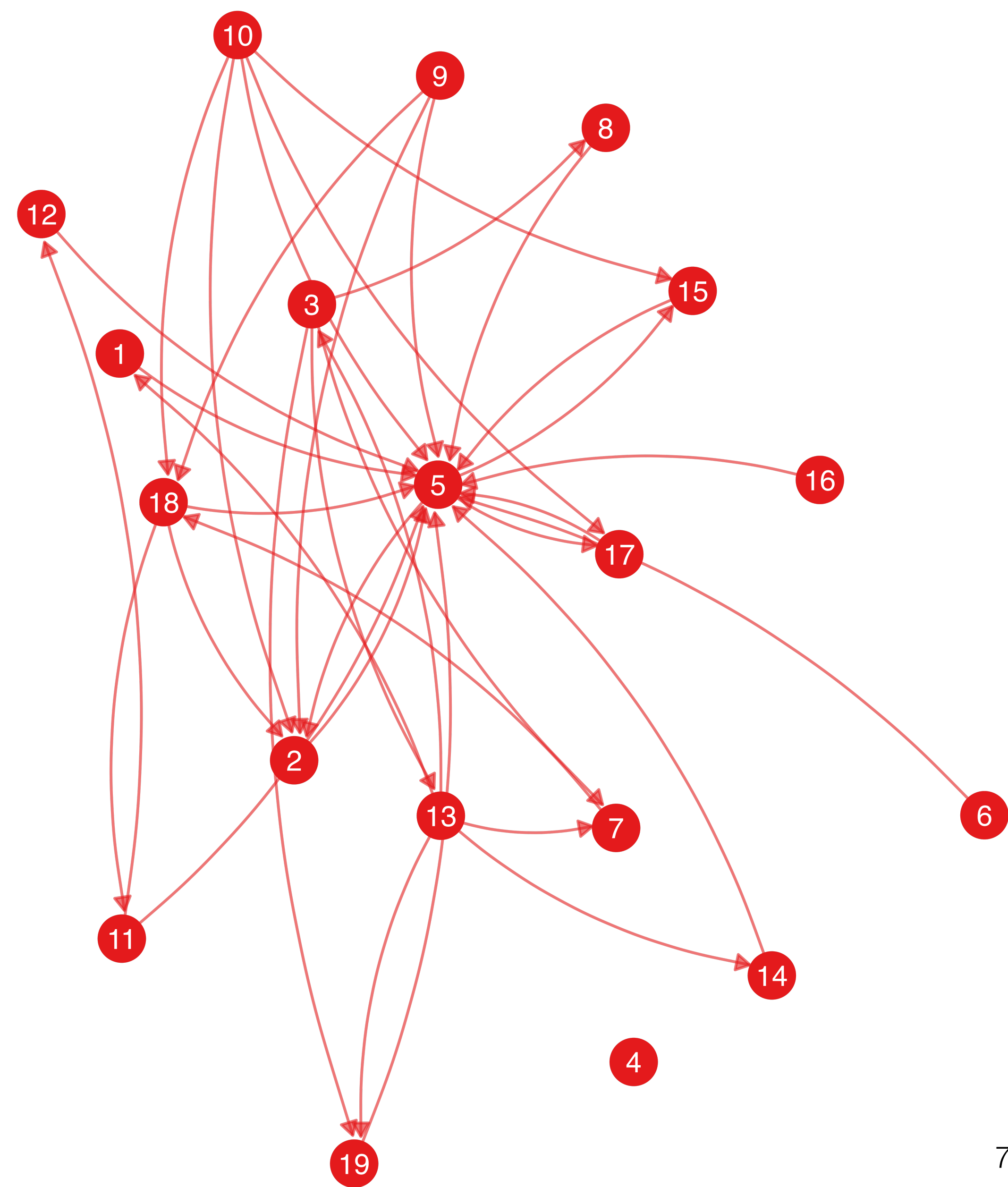
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We treat students who did not complete a given set of questions as having no nominations for that social relationship category

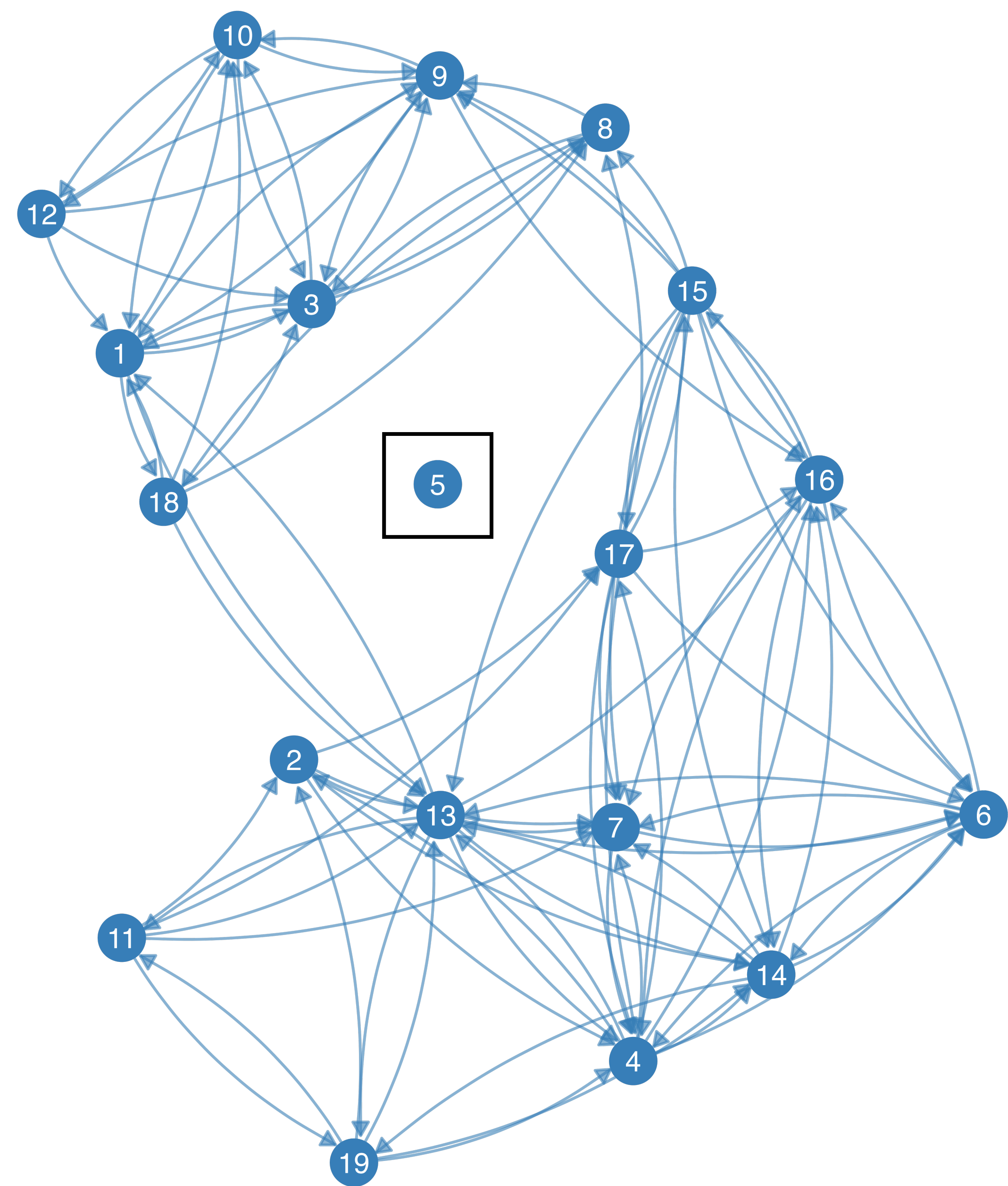
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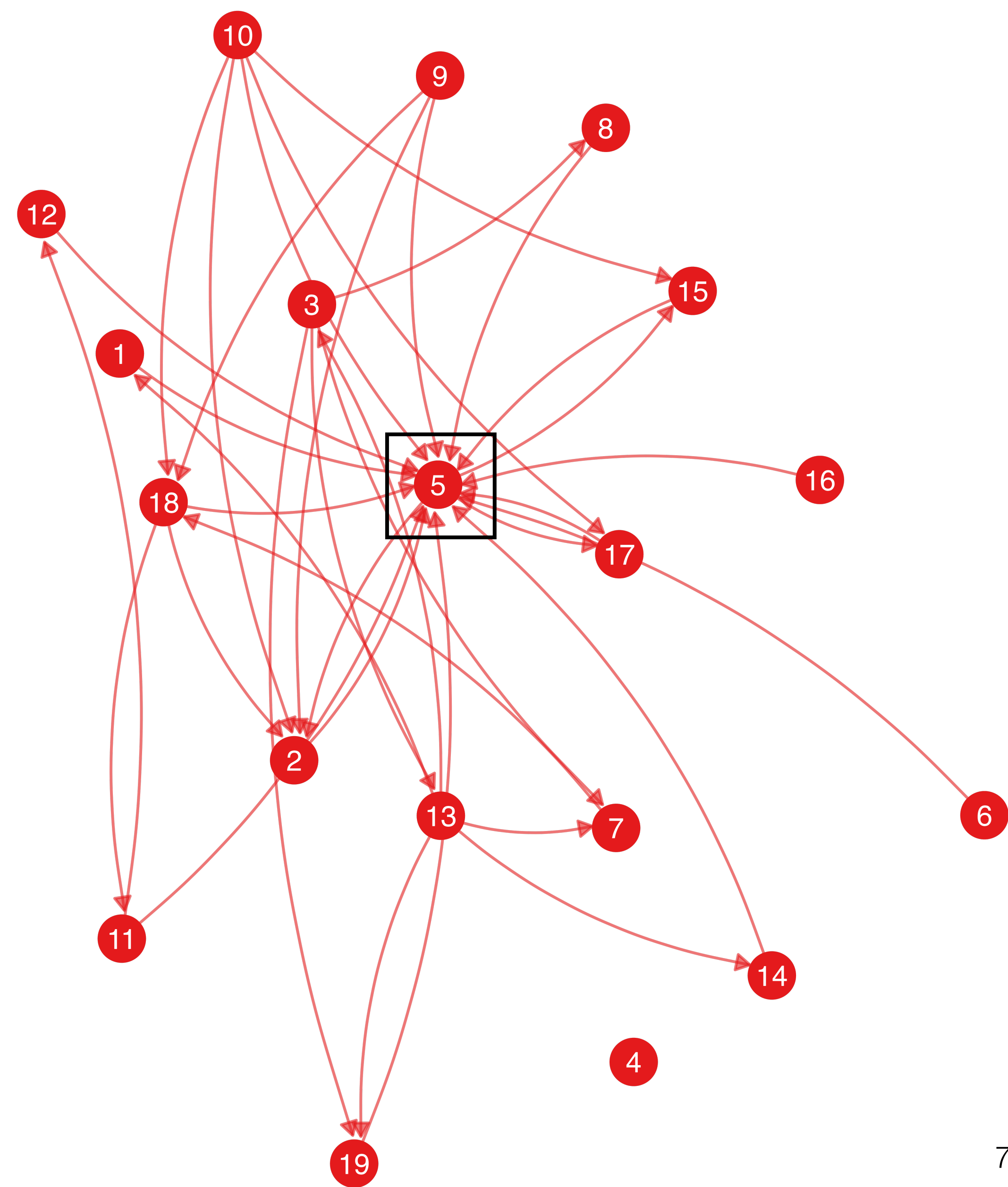
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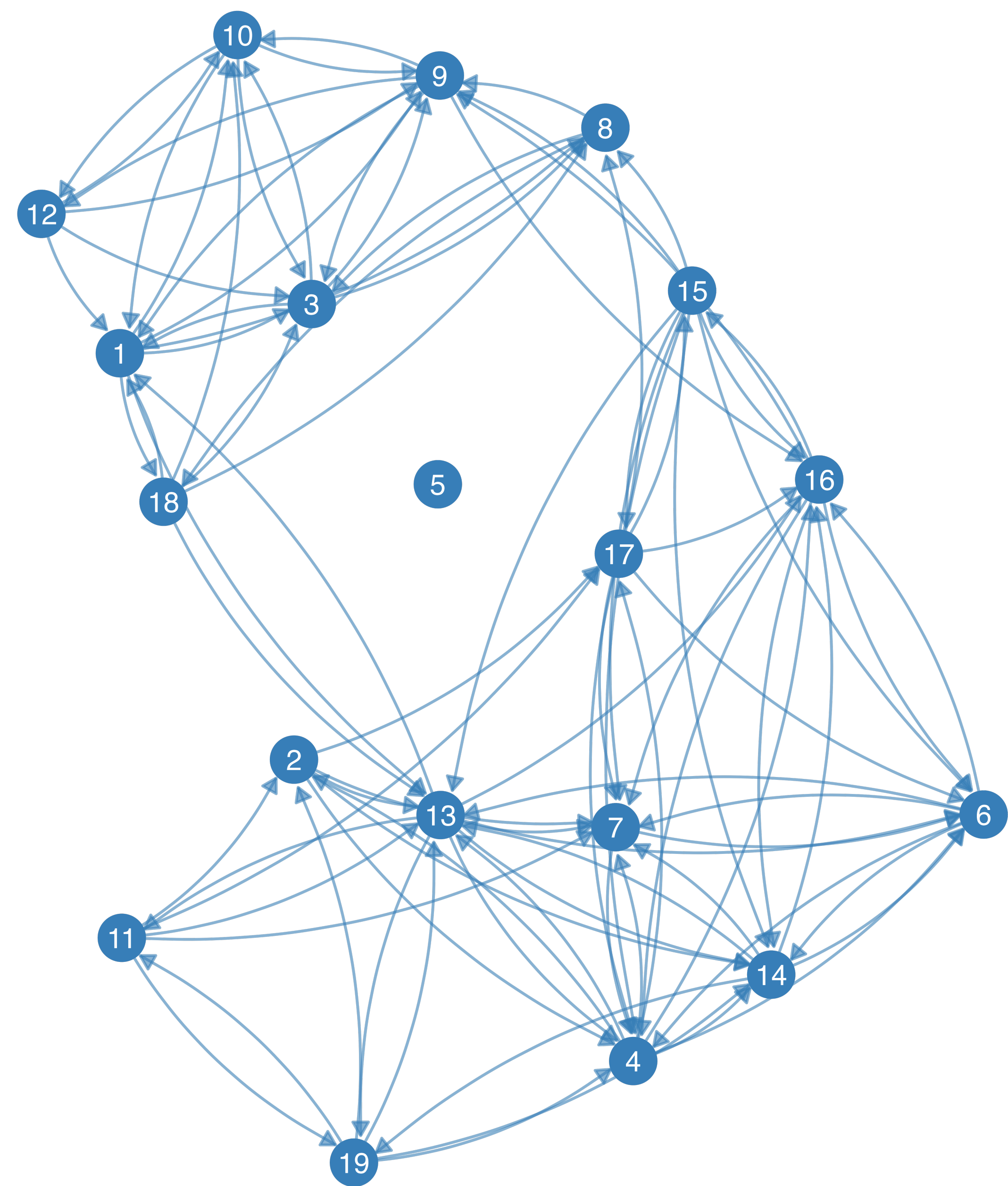
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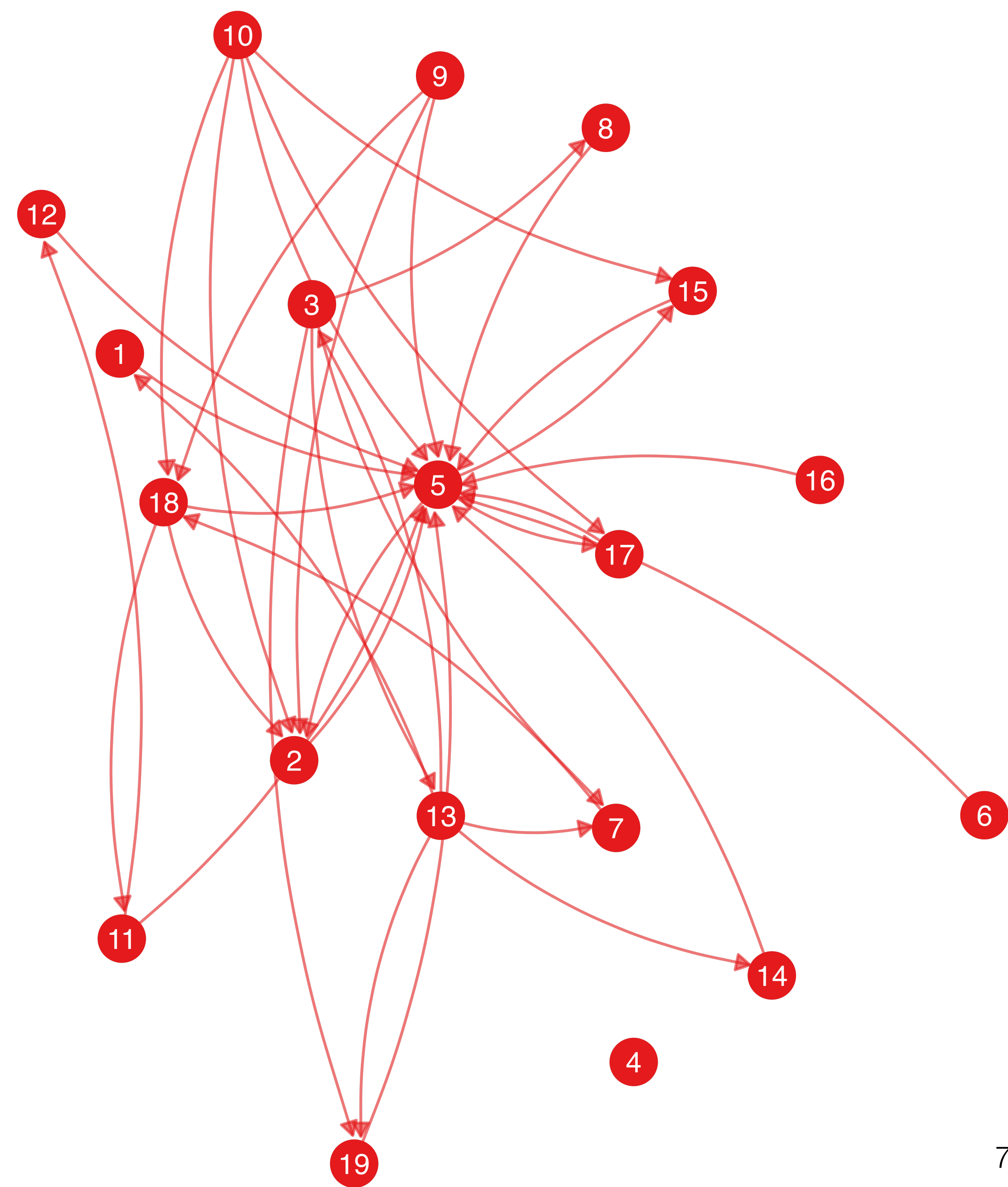
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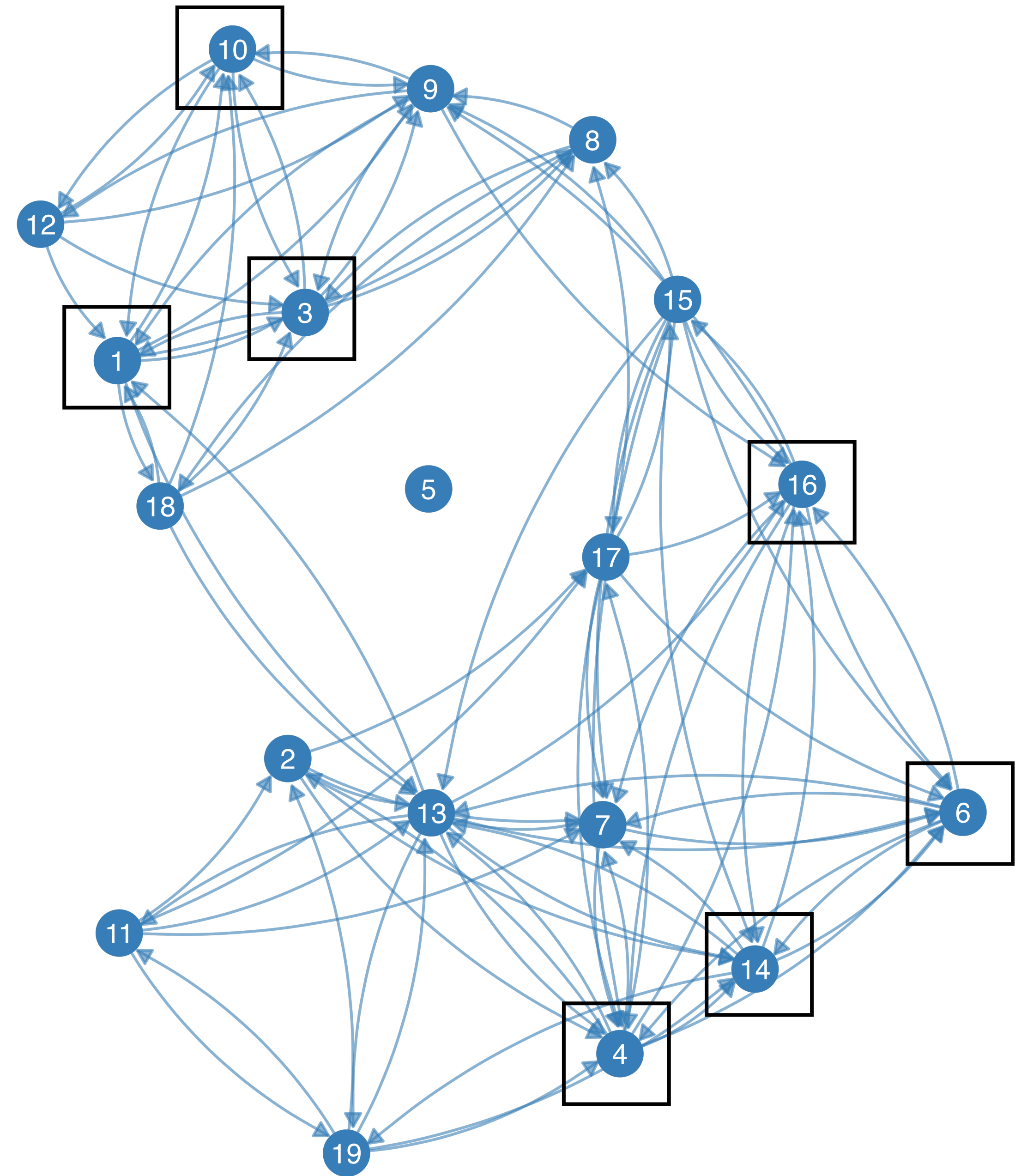
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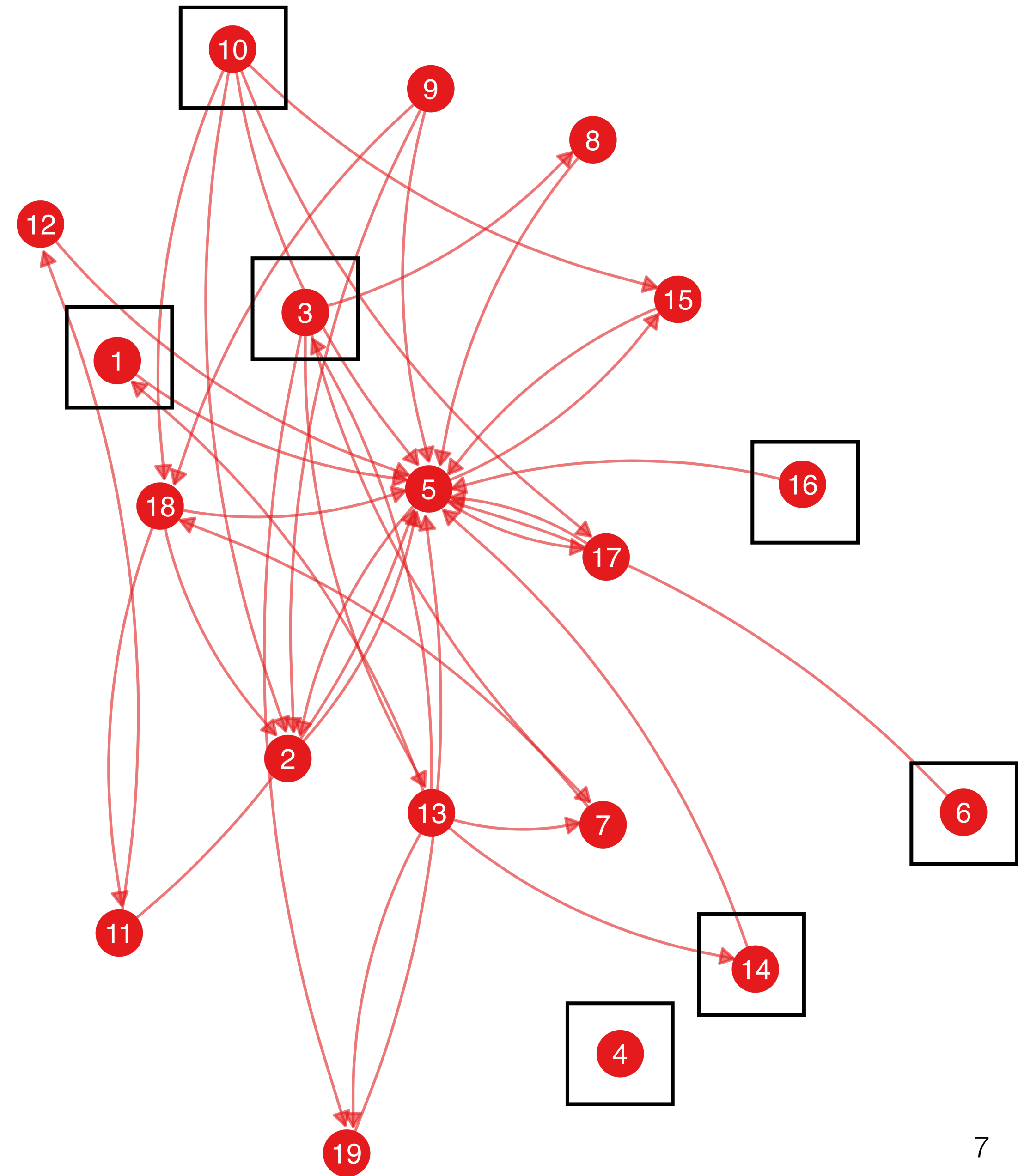
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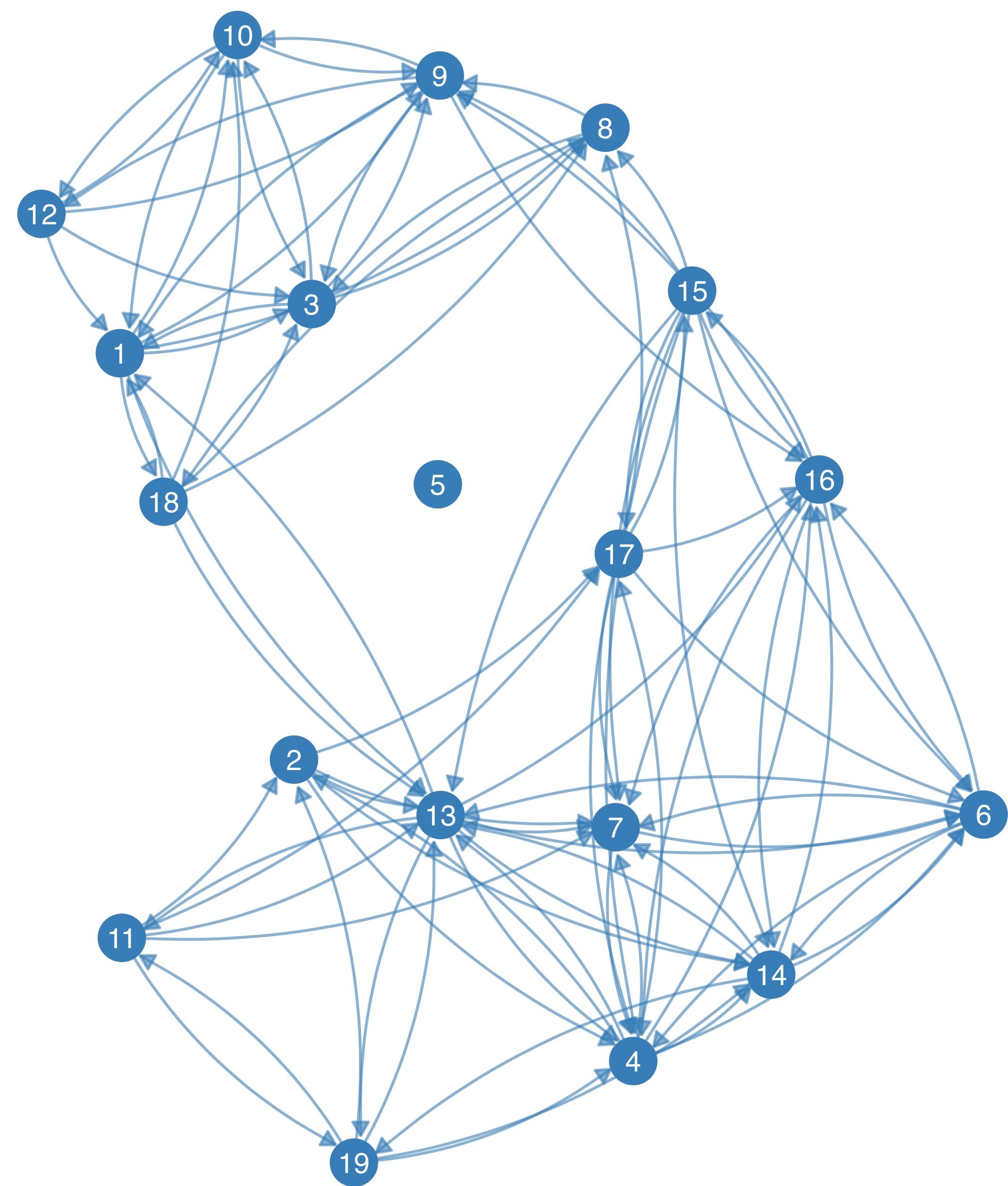
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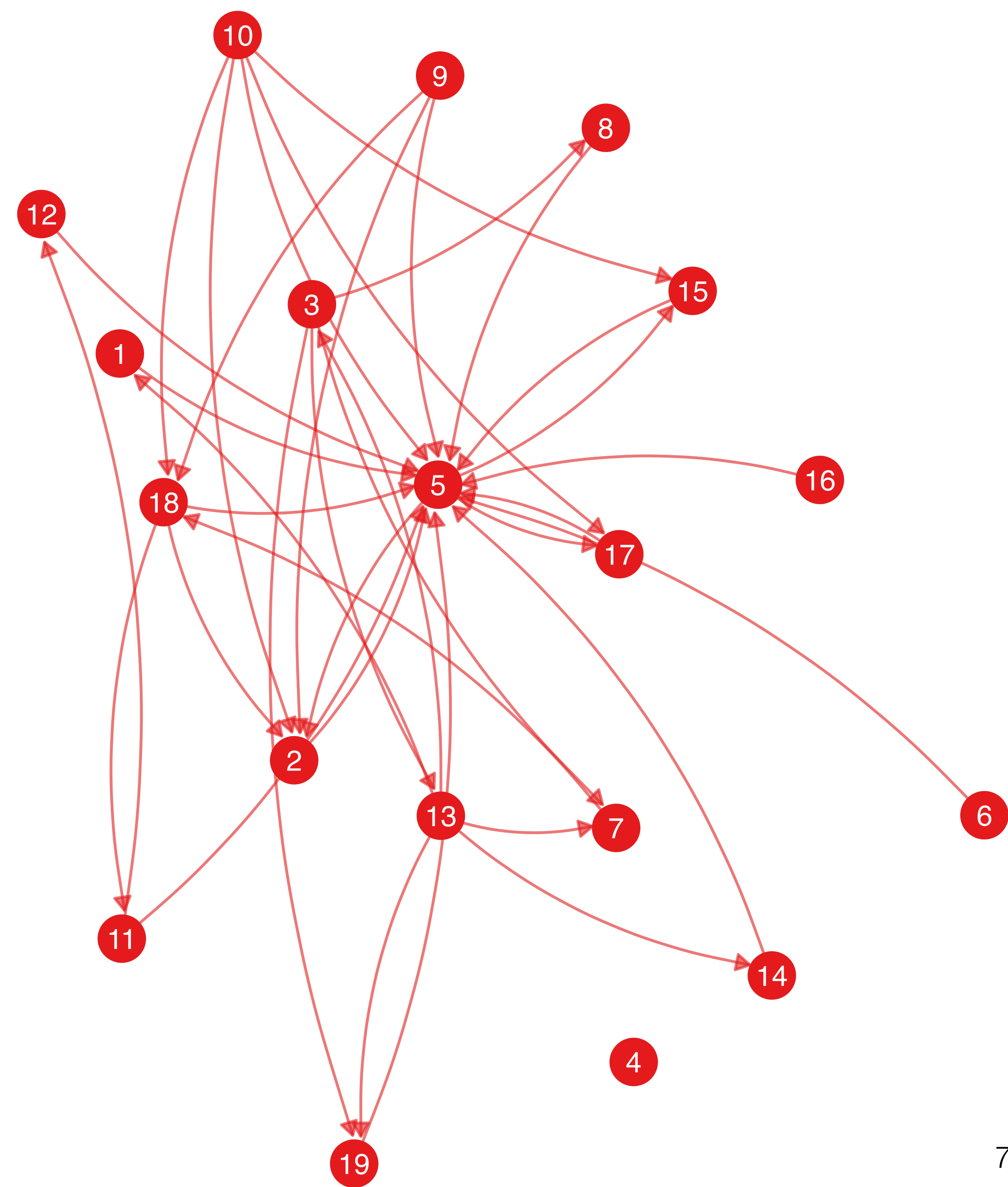
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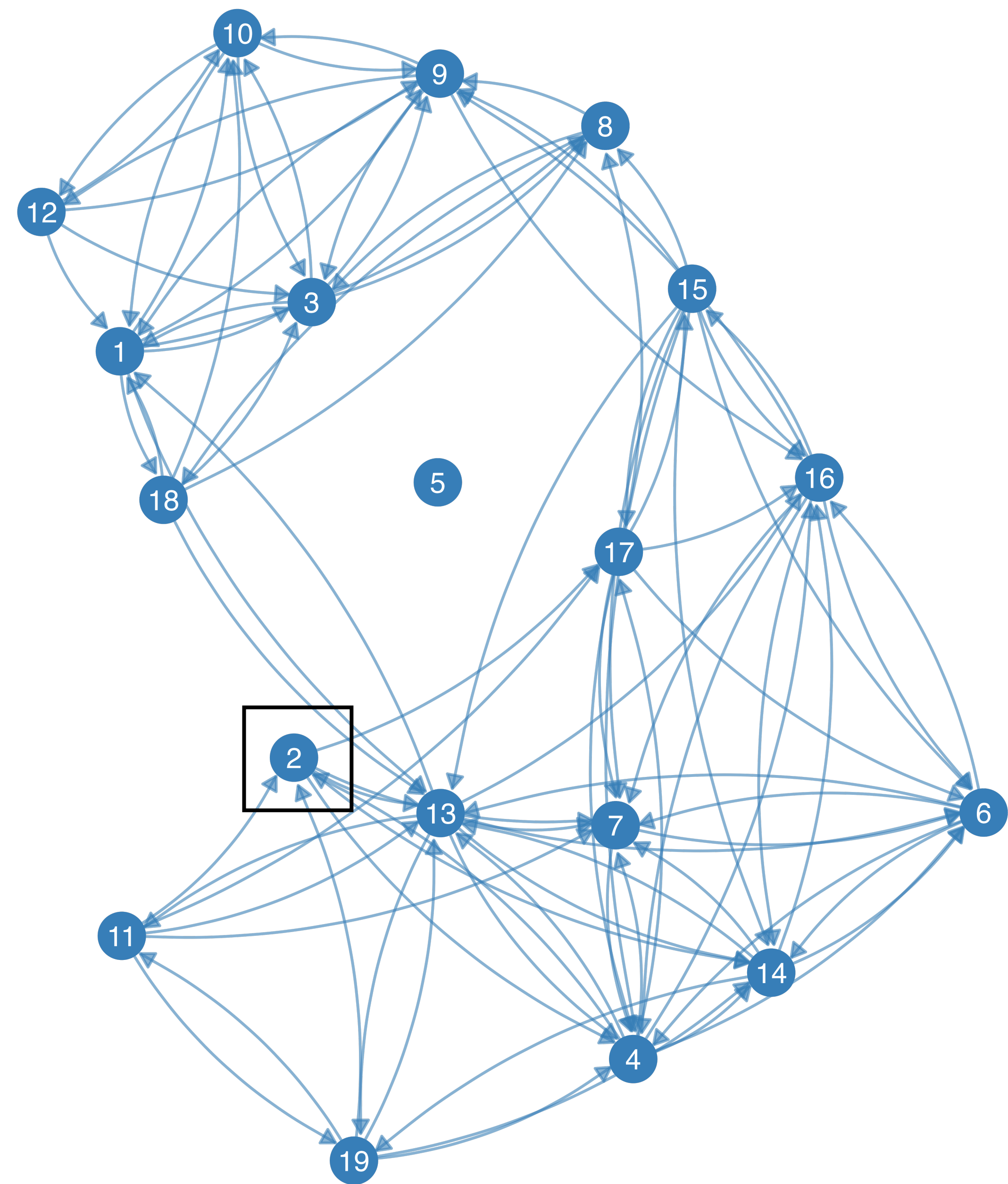
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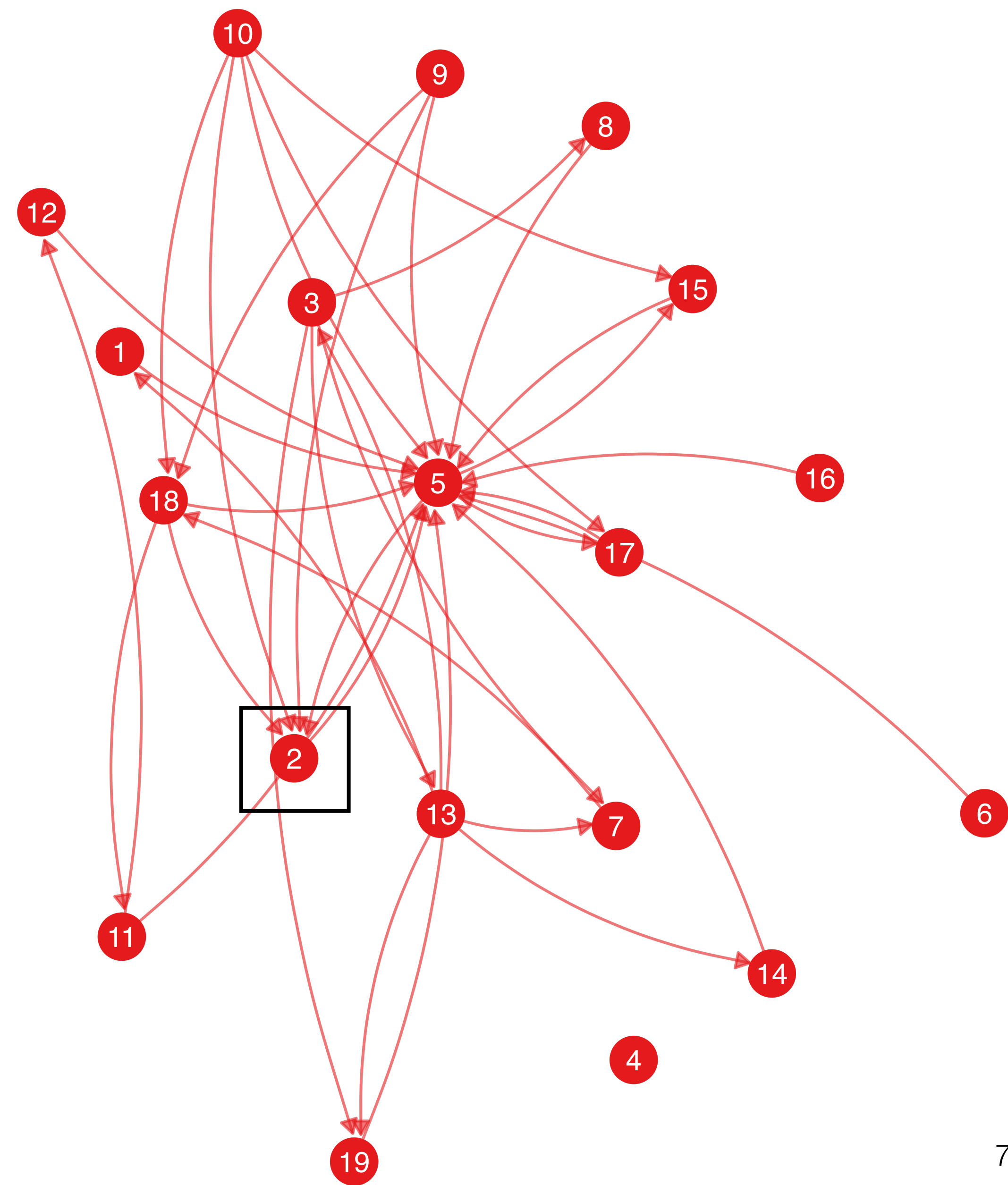
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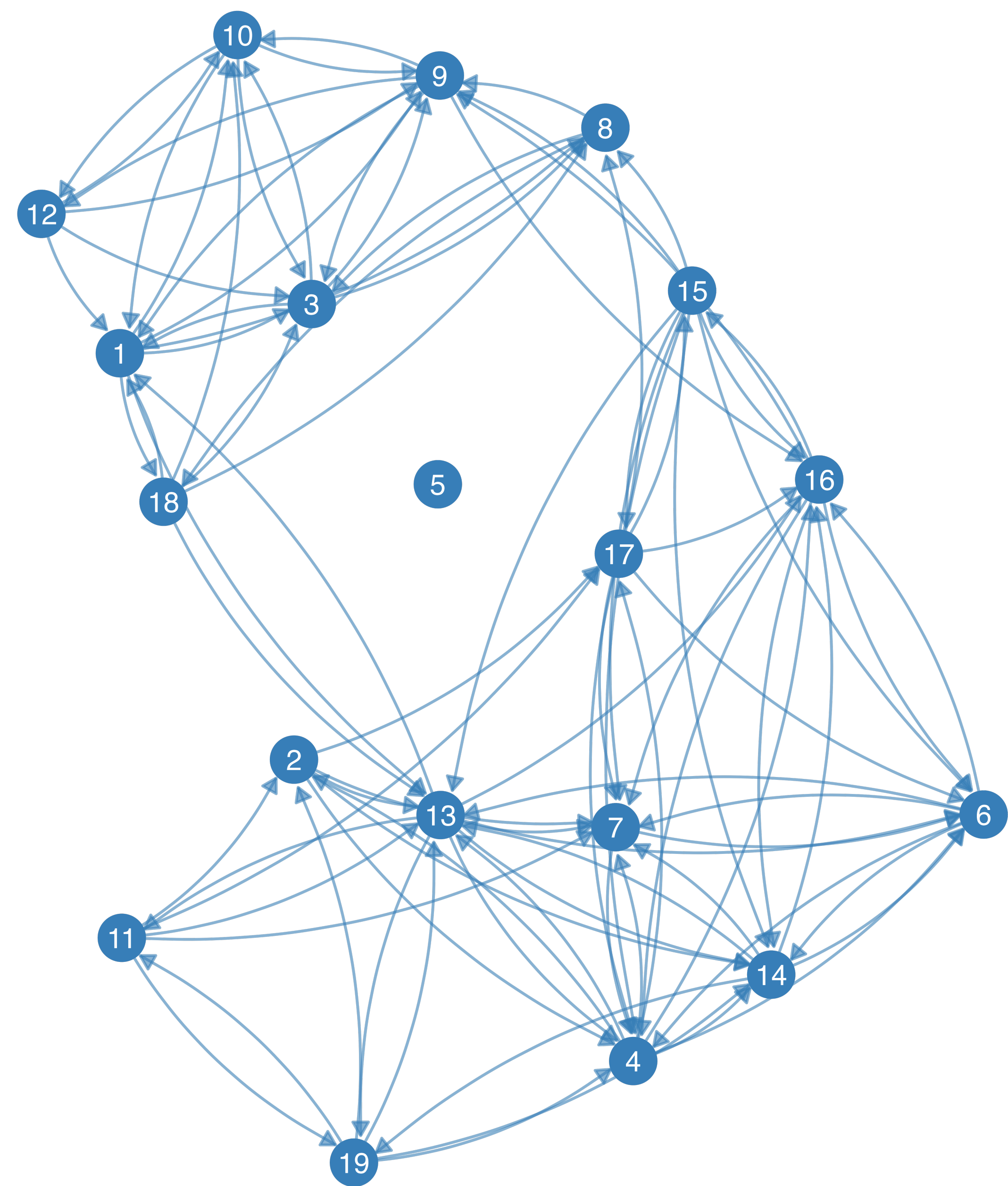
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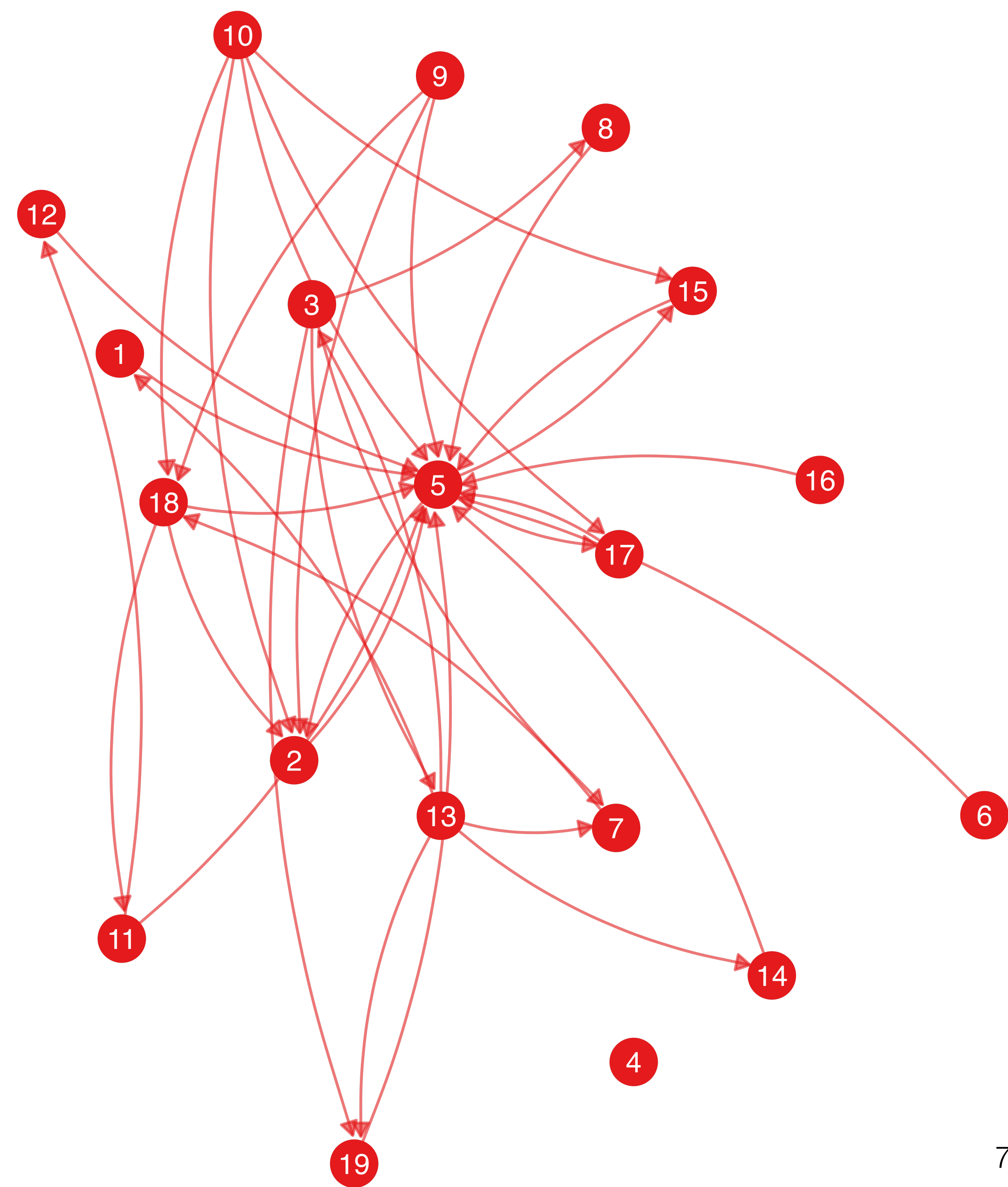
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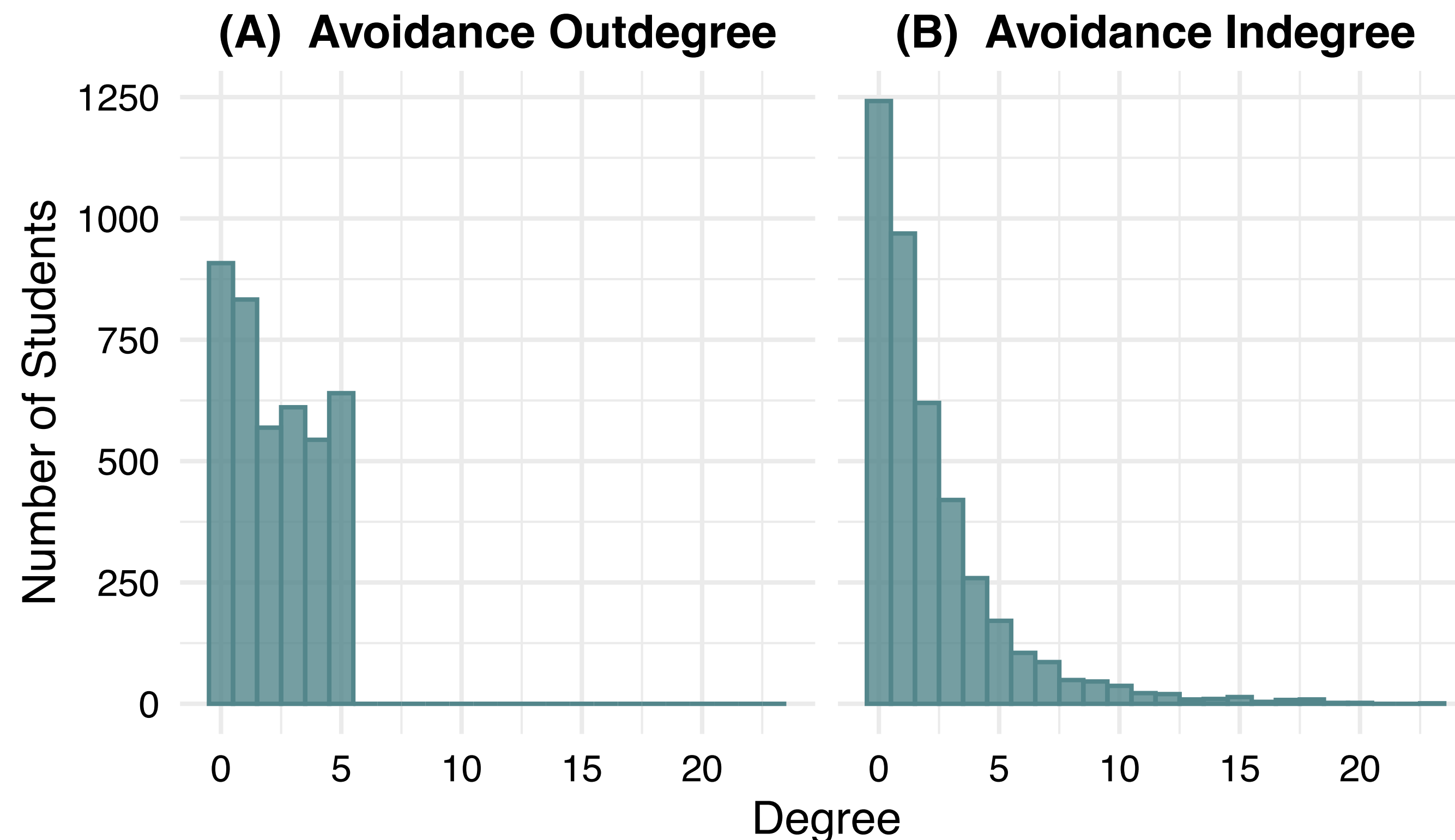
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There is little overlap between friendship and avoidance

	Mean Jaccard Index	SD Jaccard Index
Avoidance and Friendship	0.023	0.029
Avoidance and Transposed Friendship	0.027	0.026

Transposed Friendship:

If classmate j considers student i a friend $\Rightarrow$ tie from student i to classmate j

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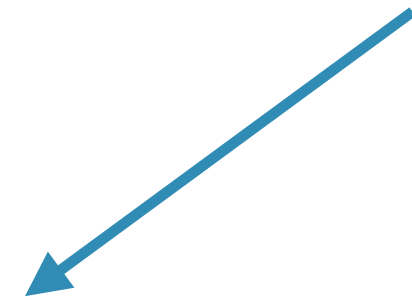
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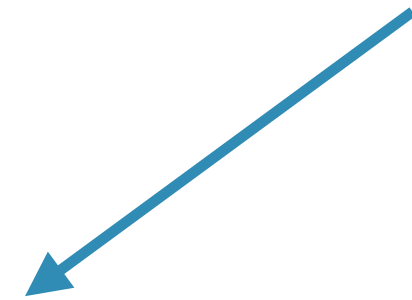
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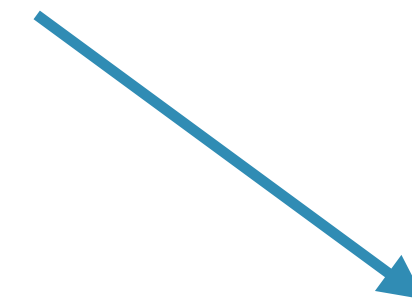
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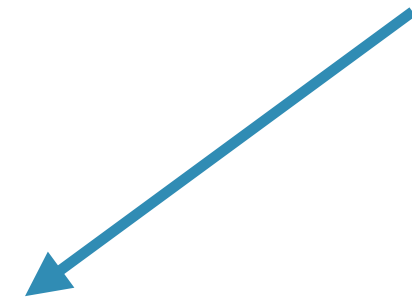
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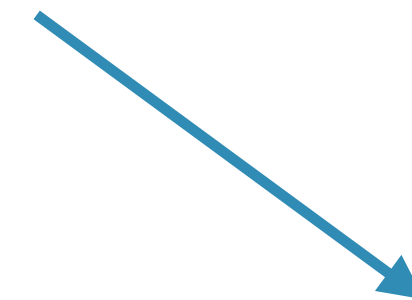
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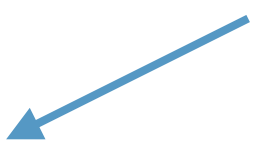
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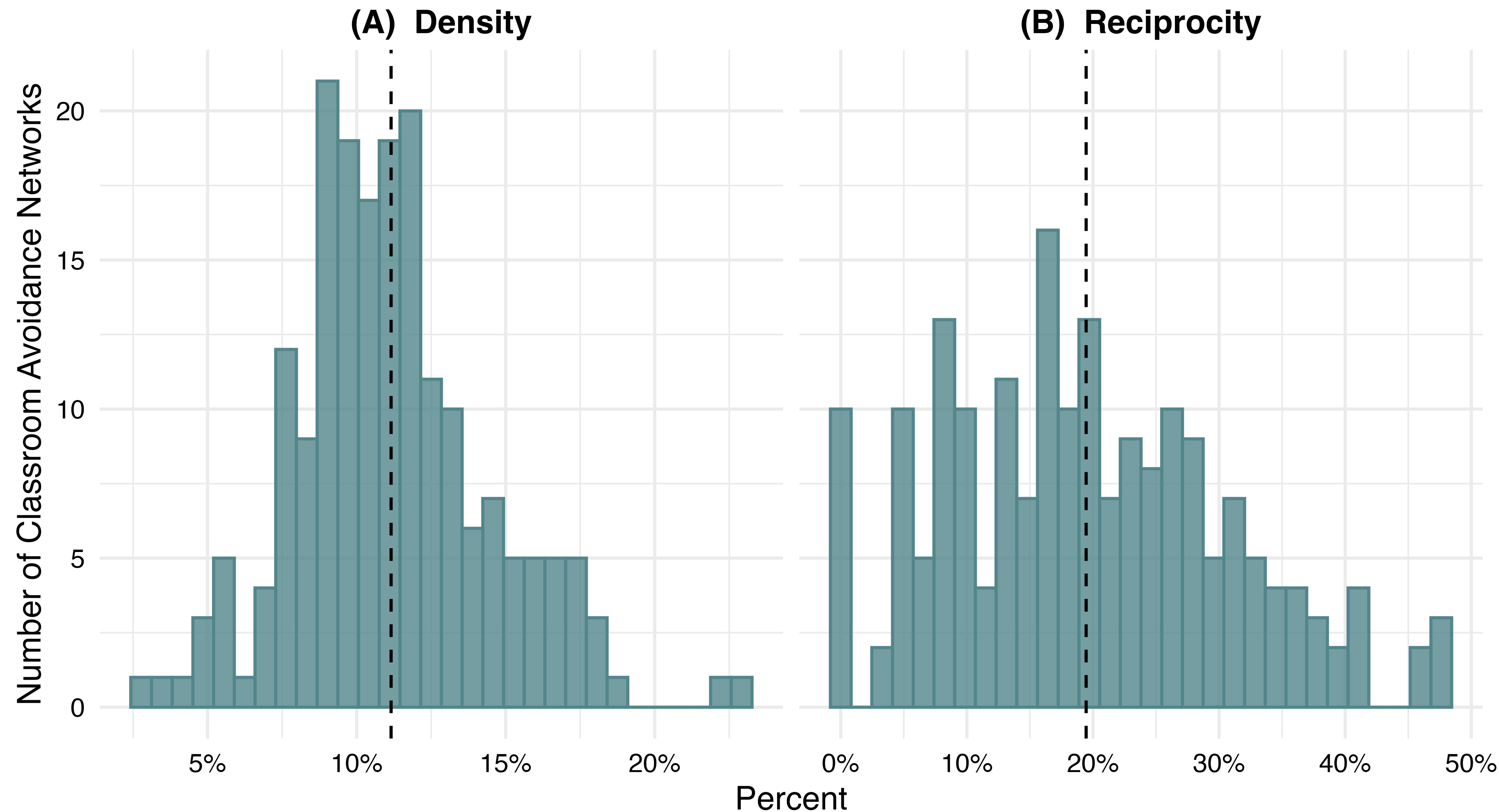
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ERGM coefficients: change in the conditional log-odds of a tie associated with a one-unit increase in the given statistic

Heterogeneity in classroom avoidance networks



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 - “Borrows strength” across networks to estimate the network-specific random effects

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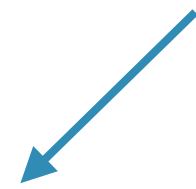


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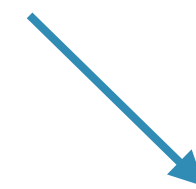
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Conditional probability of observing $y^{(k)}$ follows an exponential-family distribution:

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- $g(y^{(k)}, \mathbf{X}^{(k)})$: sufficient statistic vector for classroom k
- $\theta^{(k)}$: ERGM parameters specific to classroom k

Can contain counts of network configurations and exogenous covariates

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- $\theta^{(k)}$: ERGM parameters specific to classroom k
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We focus most of our interpretations on μ and its variance (on main diagonal of Σ)

Model specification

In our hierarchical ERGM, we include:

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Use Bayesian framework to estimate our parameters (Agneessens et al., 2024)*

*<https://github.com/johankoskinen/BayesERGM>

Hierarchical ERGM estimation

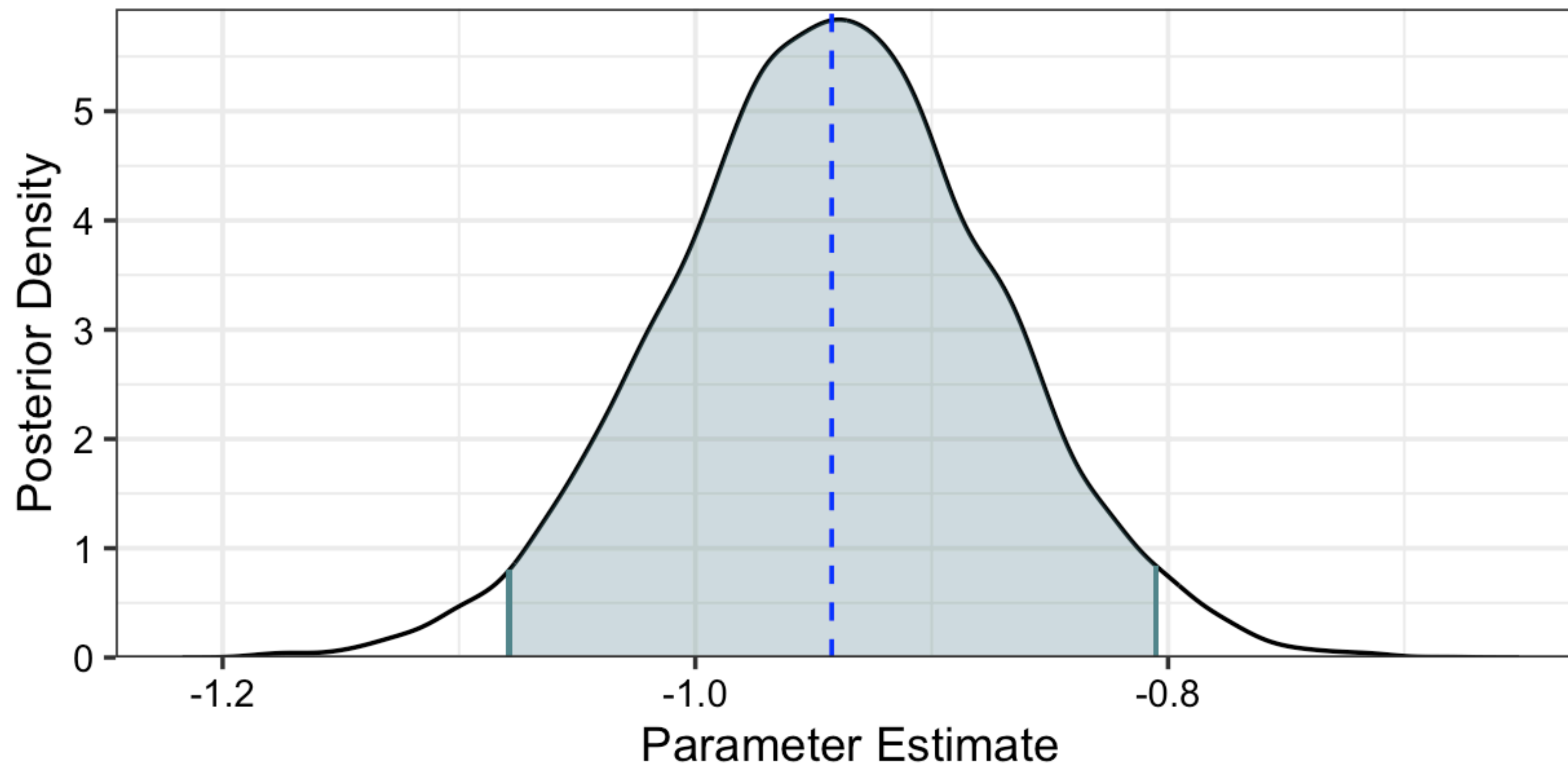
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Hierarchical ERGM estimation

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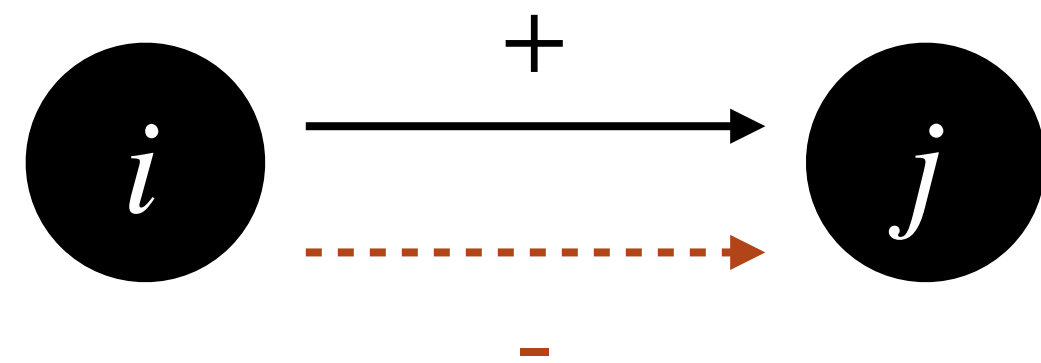
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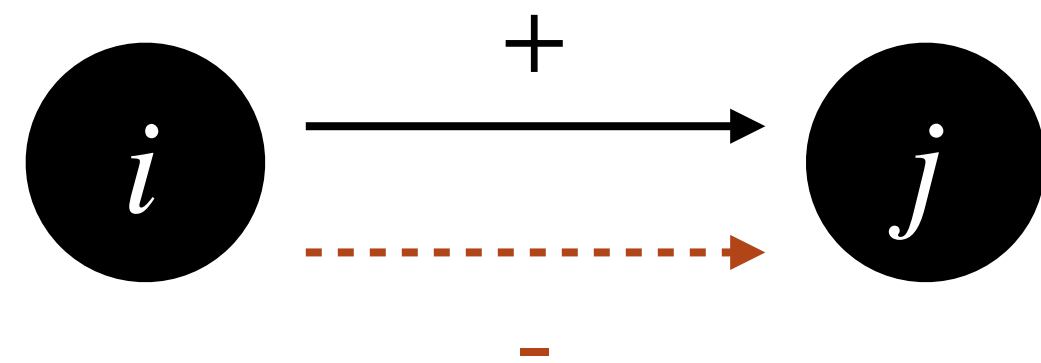
On average, avoidance networks are highly associated with friendship networks

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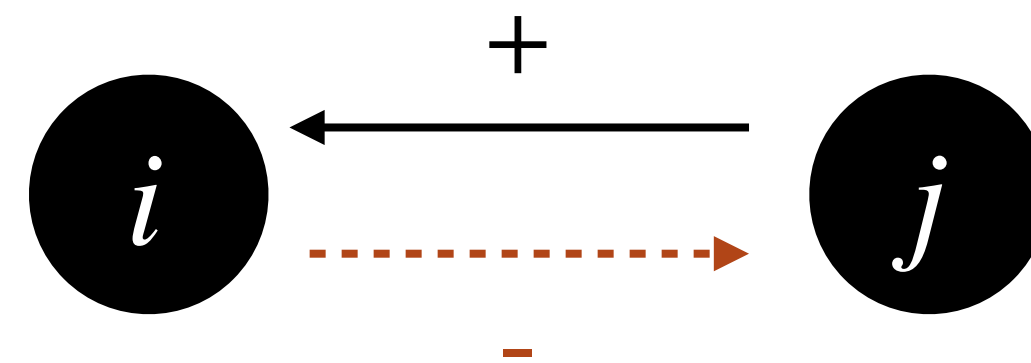


Students tend not to avoid classmates they consider a friend (OR = 0.37, 95% CI [0.32, 0.44])

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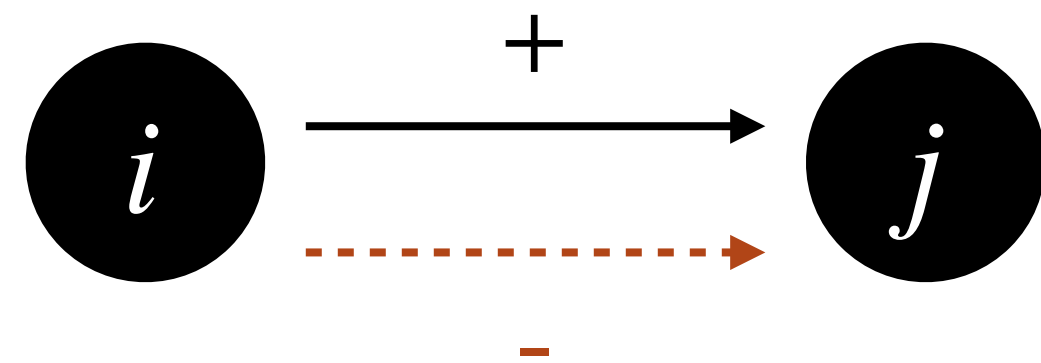


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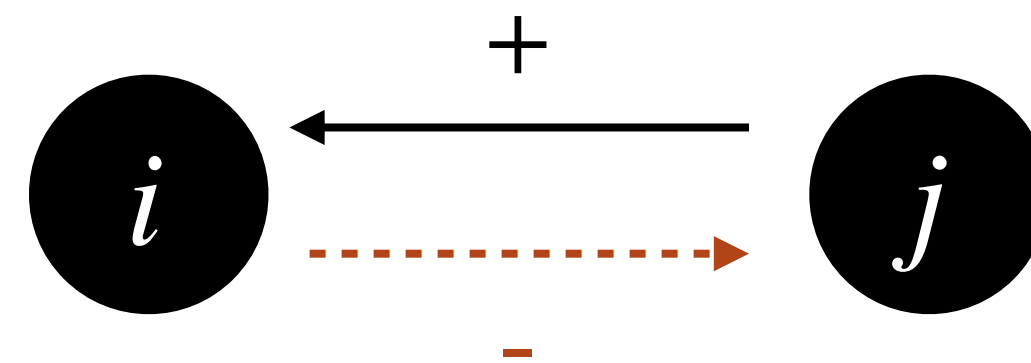


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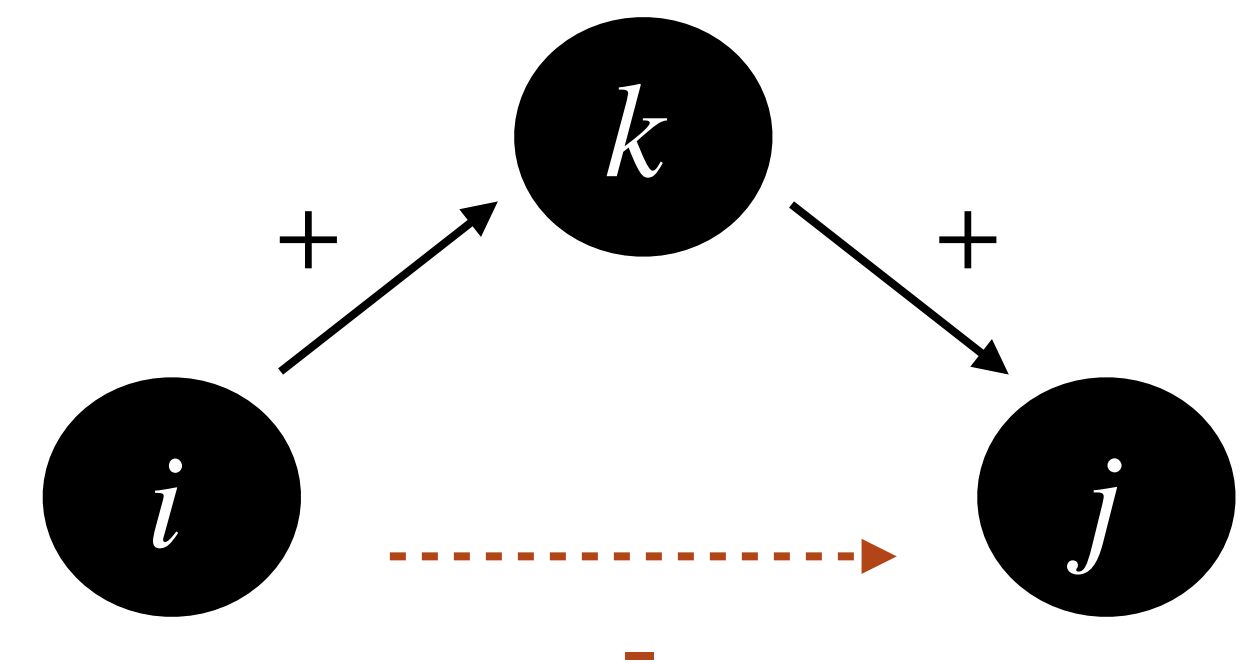
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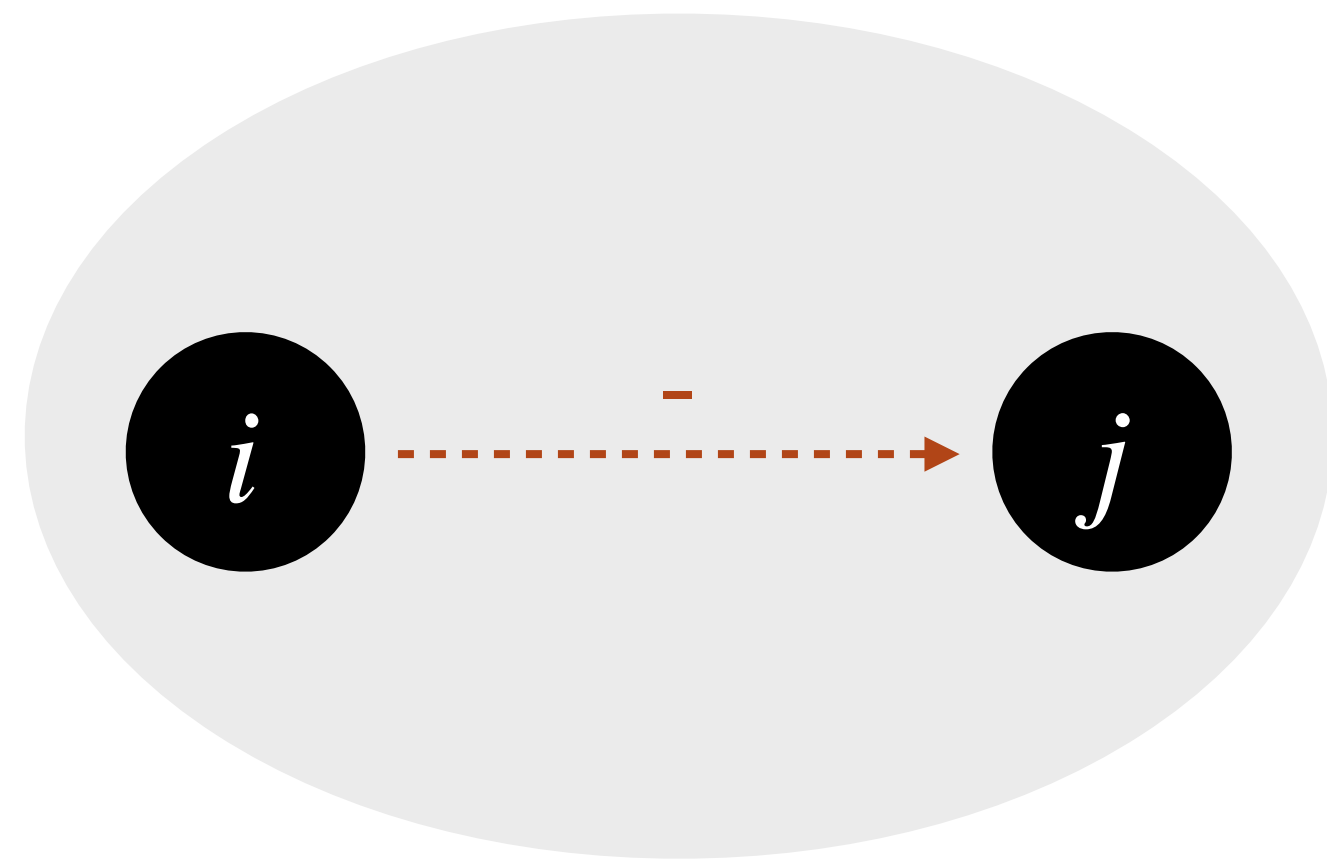
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Students tend not to avoid classmates who are "friends-of-friends" (OR = 0.75, 95% CI [0.72, 0.79])

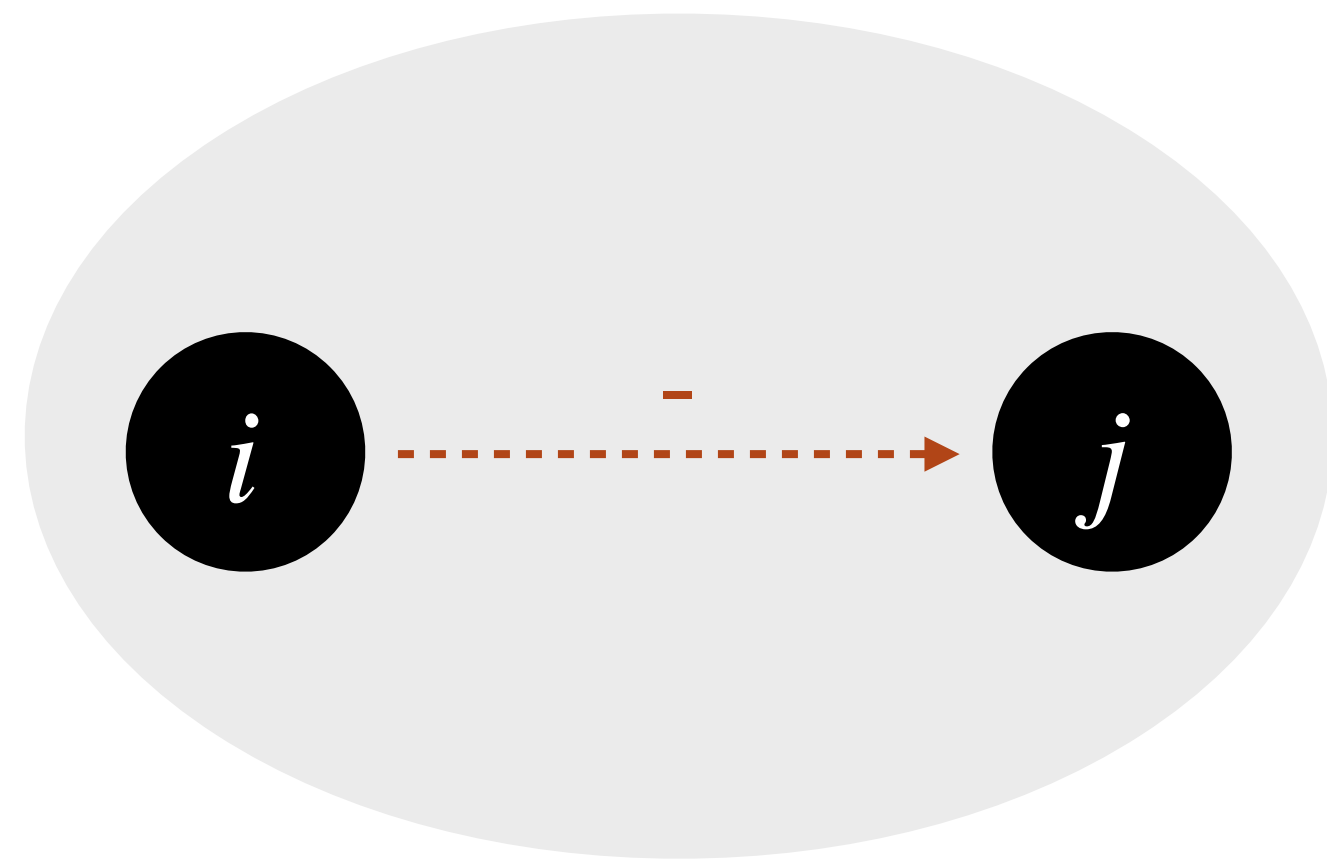
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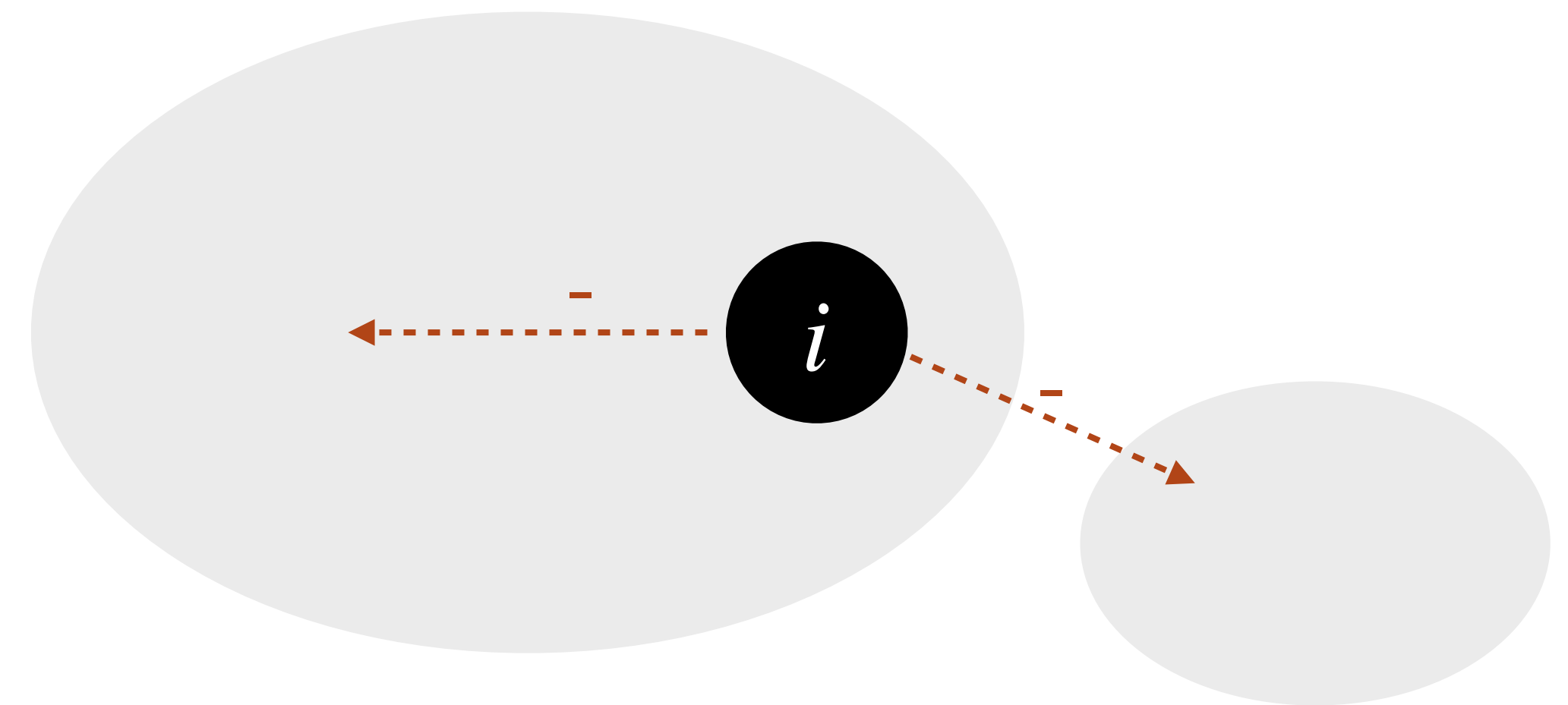


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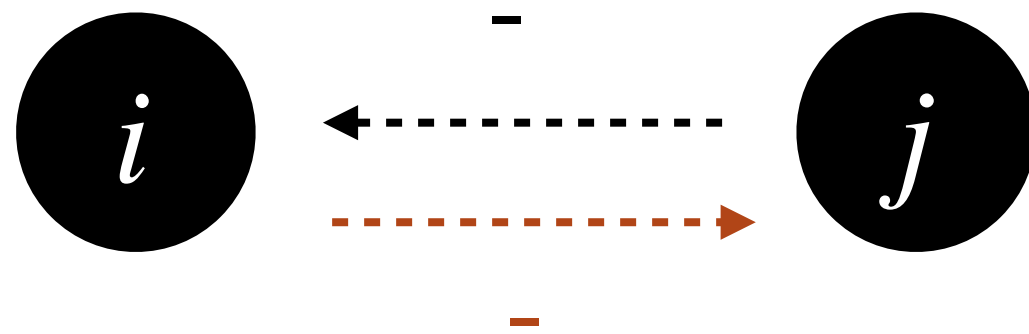
Students tend not to avoid classmates they are weakly connected to in the friendship network (OR = 0.69, 95% CI [0.63, 0.77])



Students in the largest weakly connected component of the friendship network are more likely to avoid others (OR = 1.21, 95% CI [1.09, 1.33])

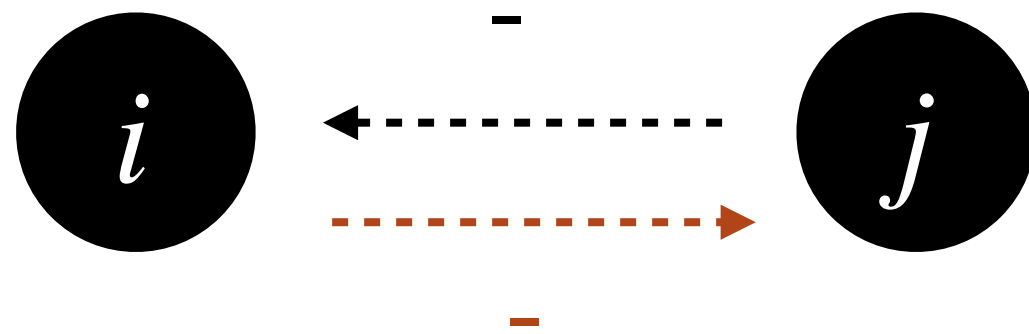
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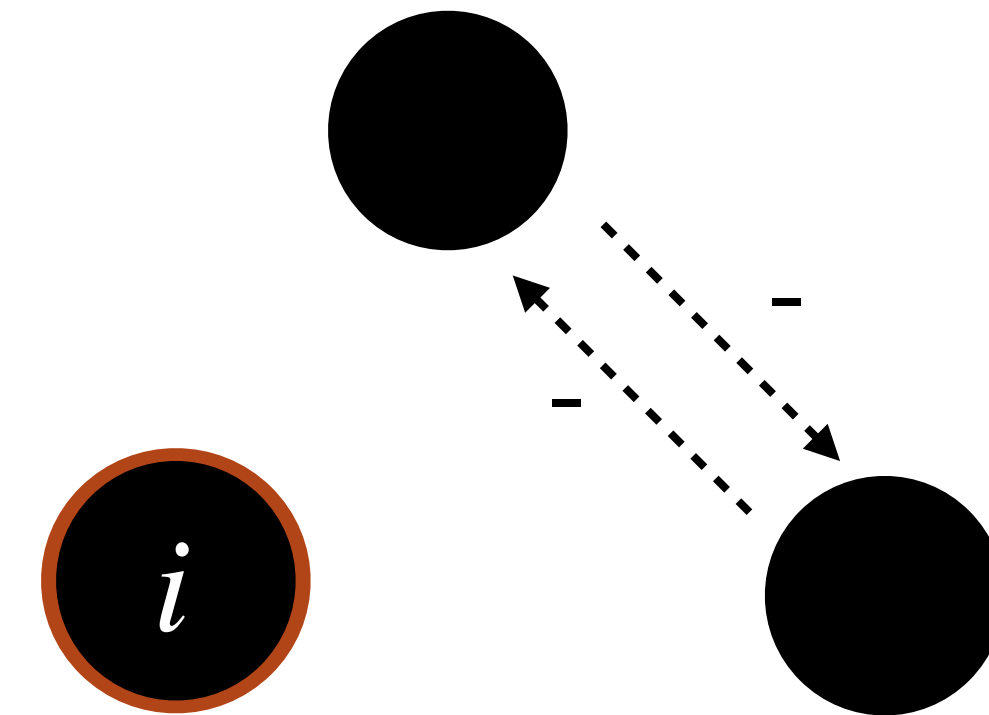


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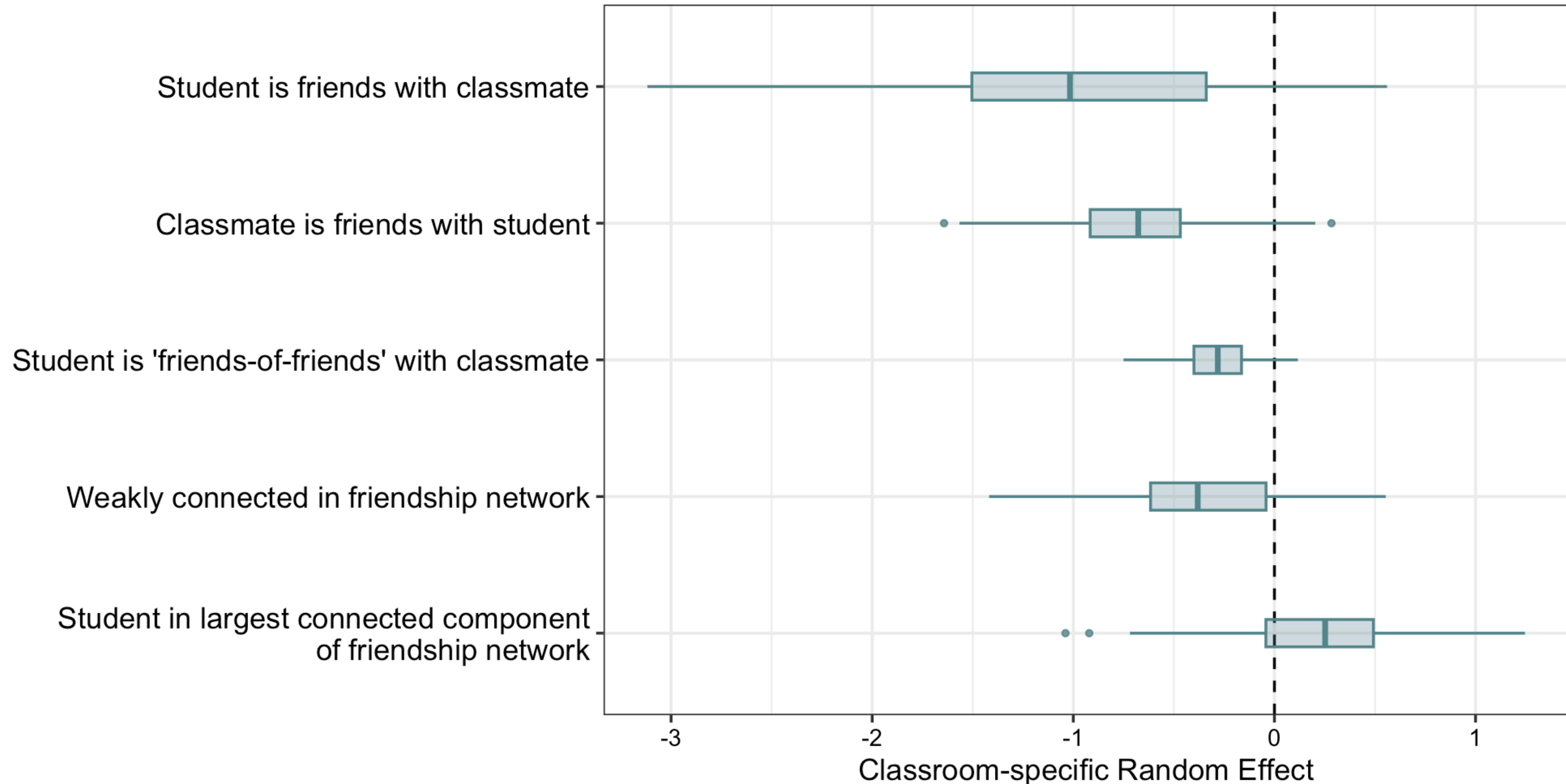


Avoidance tends to be reciprocated
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Many students avoid none of their
classmates (OR = 3.08, 95% CI [2.68, 3.54])

Variability in friendship effects across classrooms



Conclusions and future work

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Thank you! Questions?

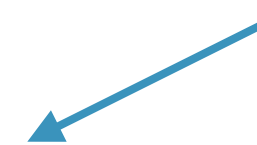
Appendix Slides

Bayesian estimation details

The Bayesian framework described by Agneessens et al. (2024) is a two-part Markov Chain Monte Carlo (MCMC) Algorithm

- (1) Update $(\theta^{(1)}, \dots, \theta^{(193)})$ via approximate exchange algorithm (Caimo and Friel, 2011)
- (2) Update (μ, Σ) via Gibbs sampling

For part (1): For each classroom $k \in \{1, \dots, 193\}$

- (1) Propose $\theta_*^{(k)}$ as update to the current parameter vector $\theta_0^{(k)}$ Set maximum outdegree to 5

- (2) Simulate a hypothetical classroom avoidance network $y_*^{(k)}$ using $\theta_*^{(k)}$
- (3) Compare sufficient statistics from $y_*^{(k)}$ with the sufficient statistics from $y^{(k)}$

We accept/reject the proposed parameter vector $\theta_*^{(k)}$ based on this comparison

Bayesian estimation details

For part (2): Place a Normal-Inverse Wishart prior on (μ, Σ) :

$$\mu \mid \Sigma \sim \text{Normal}(0, \Sigma)$$

$$\Sigma \sim \text{InvW}(\Lambda, \nu)$$

where $\nu = 1$ and $\Lambda = \text{diag}(0.1)$

Currently, we have:

- 50,000 MCMC iterations
- 40,000 iterations are discarded (burn-in) and thinning factor of one

Last 10,000 samples are used to estimate the hierarchical ERGM parameters